

Database Project: Human Resources Management System (HRMS)

Team Members:

- Sarah Elaraby Aboayad
- Semon Mamdouh Beskales
- Ahmed Adel Tawfik
- Mohamed Essam Mahrous

1. Project Overview:

This project aims to design and implement a comprehensive Human Resources Management System (HRMS) using Microsoft SQL Server. The system manages employee data, departments, job roles, salaries, and leaves. It also incorporates advanced SQL techniques to ensure efficiency and flexibility.

2. Project Phases:

A. Analysis & Design:

Responsible Members: Sarah Elaraby, Simon Mamdouh

- Designed the Entity Relationship Diagram (ERD).
- Created the entity-to-table mapping.
- Defined primary and foreign keys for table relationships.

B. Database Creation & Implementation:

Responsible Members: Ahmed Adel, Mohamed Essam

- Created database tables using DDL (Data Definition Language).
- Executed data operations using DML (Data Manipulation Language).
- Queried data using DQL (Data Query Language).
- Implemented indexes to optimize performance.
- Built views to display customized sets of data.

C. Advanced Features & Coding:

- **Mohamed Essam**

- Developed user-defined functions.
- Implemented window functions for analytical queries.
- Created rules for input validation.
- Set default values for specific columns.

- **Ahmed Adel**

- Created views.
- Worked with DDL, DML, and DQL commands.
- Implemented indexes for performance tuning.

- **Semon Mamdouh**

- Used the MERGE statement for data merging.
- Implemented cursors for row-by-row operations.
- Wrote sub-queries for nested data retrieval.
- Applied UNION to combine result sets.
- Used ROLLUP and CUBE for multi-level aggregation.
- Implemented GROUPING SETS for advanced grouping.

- **Sarah Elaraby**

- Created stored procedures for modular operations.
- Used various types of joins.
- Applied aggregate functions for summary data.
- Implemented triggers for automatic actions on data changes.

3. Tools & Technologies Used:

- Database Engine: Microsoft SQL Server
- Query Language: T-SQL (Transact-SQL)
- Design Tools: ERD tools like dbdiagram.io

4. Challenges Faced:

- Coordinating tasks and responsibilities among team members.
- Handling complex relationships and constraints.
- Implementing and testing advanced SQL features.

5. Conclusion:

The team successfully designed and implemented a fully functional HR management system using SQL Server. The project enhanced our technical skills and improved our collaboration as a team. Each member played a vital role in bringing the project to completion.